

Matching data sets to project tasks

Looking at your datasets and tasks, here's the mapping: Here's a clear breakdown of which dataset fits each task:

Dataset → Task Mapping

Task	Level	Dataset	Why
Task 1: Data Cleaning & Preprocessing	L1	✔ Stock Prices (you chose this)	Has dates, multiple symbols, potential duplicates/inconsistencies — good for cleaning
Task 2: EDA	L1	Iris	Classic EDA dataset — clean numerics, a categorical target, perfect for distributions, correlations, boxplots
Task 3: Basic Data Visualization	L1	Iris or House Prices	Both work; Iris is simpler for bar/scatter plots
Task 1: Regression Analysis	L2	House Prices	Continuous target variable (price), multiple numeric features — textbook regression use case
Task 2: Time Series Analysis	L2	Stock Prices	Has a <code>date</code> column and price data — ideal for trend/seasonality decomposition
Task 3: Clustering (K-Means)	L2	Iris	Well-known clustering benchmark; 3 species map naturally to 3 clusters
Task 1: Predictive Modeling (Classification)	L3	Churn (bigml-80 for train, bigml-20 for test)	Has a binary <code>Churn</code> column, pre-split into 80/20 — purpose-built for classification
Task 2: Dashboards	L3	Any (Stock or Churn recommended)	Stock prices give good time-based visuals; Churn gives category breakdowns
Task 3: NLP Sentiment Analysis	L3	Sentiment dataset	Has a <code>Text</code> column and a <code>Sentiment</code> label — exactly what's needed

Key callouts:

- The **churn files are pre-split** (80% train / 20% test) — use both together for the classification task, not separately.
- The **house prices dataset has no headers**, so you'll need to assign column names manually (it's the Boston Housing dataset format).
- **Iris** is versatile enough to appear in both EDA and Clustering tasks without conflict since they're different levels.